# Where $\alpha$ hides

Charge unit, two spheres and $\xi$ as an area

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# Where $\alpha$ hides

**Abstract**

FFGFT is first and foremost a ratio-based theory. It works in natural units with the convention $\alpha = 1$ and a correspondingly redefined charge; in this basic form it needs no fractal correction. This document shows that nothing is lost in doing so and that the geometry becomes visible. With $\hbar = c = \varepsilon_0 = 1$ one has $e^2 = 4\pi\alpha$; with $\alpha = 1$, $e = \sqrt{4\pi}$, and the charge is measured by the full surface of the sphere. $\alpha$ moves into the charge unit and is the ratio of two spheres around every charged particle, $\alpha = r_e/\lambda_C$; with $\alpha = 1$ the Coulomb sphere, the Compton sphere and the Bohr radius merge into one scale. In the basic form the FFGFT bridge reads $\xi E_0^2 = 1$ with $E_0^2 = m_e m_\mu$. This has an immediate geometric reading: $\xi$ is the product of the Compton radii of electron and muon, $\xi = \lambda_e \lambda_\mu$, an area. The number $1/\xi = 7500 = 100 \cdot 75$ is the same one that carries the recursion, with starting value $\xi_0 = 1/7500$ and rotation number $74/75$. Only the translation into SI needs a reference energy, the MeV, and there, and only there, the fractal correction appears.

## Status markers

**[B]** Algebraically proven.
**[K]** Numerically confirmed (check script, comparison with measured values).
**[S]** Assumption, open or speculative.
**[X]** Refuted or not sustainable.
**[Q]** Convention.

Constants: CODATA 2022. Check script: `pruef_386_alpha_geometrie.py`.

# 1 The basic form of FFGFT

FFGFT calculates with ratios. In natural units, mass, energy, temperature and inverse time are one quantity in four guises (A085), and the coupling is adopted as a unit of calculation: $\alpha = 1$ **[Q]** (A267). The only parameter is $\xi = 4/30000 = 1/7500$. In this basic form there is no fractal correction; $K_{\text{frak}}$, the SI values of $\alpha$ and $G$ and the fully worked-out mass tables belong to the translation into SI, not to the core (A130). The question of this document is therefore not how precisely $1/137$ is reproduced, but what $\alpha = 1$ means geometrically and which structure the basic form then shows.

# 2 $\alpha = 1$: the charge as the full sphere

The fine-structure constant is

$$\alpha = \frac{e^2}{4\pi\varepsilon_0 \hbar c}. \tag{1}$$

In natural units with $\hbar = c = \varepsilon_0 = 1$ (Heaviside–Lorentz) what remains is

$$e^2 = 4\pi\alpha \quad \textbf{[B]}. \tag{2}$$

The $4\pi$ is the full surface of the sphere. By Gauss's law the charge is the flux through a closed sphere; the flux is spread over the solid angle $4\pi$. Heaviside–Lorentz puts the $4\pi$ exactly where it belongs geometrically, into Coulomb's law, and keeps the Maxwell equations free of it (A267). Setting $\alpha = 1$ gives

$$e_{\text{geo}} = \sqrt{4\pi} = 3.545, \tag{3}$$

so the charge is measured geometrically: its square is the full solid angle. With the measured $\alpha$ the historical charge is $e_{\text{hist}} = \sqrt{4\pi\alpha} = 0.3028$. Hence

$$\alpha = \left(\frac{e_{\text{hist}}}{e_{\text{geo}}}\right)^2, \qquad \frac{e_{\text{geo}}}{e_{\text{hist}}} = \frac{1}{\sqrt{\alpha}} = 11.706 \quad \textbf{[B]}. \tag{4}$$

Under the redefinition $\alpha$ therefore does not disappear. It moves into the charge unit: $\alpha$ is the square of the ratio of the historical to the geometric charge unit. The calculation stays fully reversible, because $e^2 = 4\pi\alpha$ carries the coupling order along: one factor $\sqrt{\alpha}$ per vertex, and a rate of order $\alpha^n$ gets exactly the factor $\alpha^n$ back in the transition to SI (A267).

The same $4\pi$ appears in the Higgs–vacuum route: the factor $64\pi^4 = 16\pi^3 \cdot 4\pi$ of the 2025 vacuum formula is the $4\pi$ from $4\pi\varepsilon_0$; with $\alpha = 1$ the vacuum formula becomes the Higgs expression that checks $\xi$ to about $2.3\,\%$ (Doc. 385). The basic form thus connects the geometry of the charge with the check of $\xi$ from the Higgs sector.

## 3 Two spheres around every charged particle

Two spheres can be drawn around the electron. On the first, the Coulomb energy equals the rest energy; this is the classical electron radius

$$r_e = \frac{e^2}{4\pi\varepsilon_0 m_e c^2} = 2.818 \times 10^{-15}\ \text{m}. \tag{5}$$

On the second, the quantum energy $\hbar c/r$ equals the rest energy; this is the reduced Compton wavelength

$$\lambda_C = \frac{\hbar}{m_e c} = 3.862 \times 10^{-13}\ \text{m}. \tag{6}$$

Their ratio is exactly the fine-structure constant, and the Bohr radius follows with the inverse power:

$$r_e : \lambda_C : a_0 = \alpha : 1 : \alpha^{-1} \quad \textbf{[B]}. \tag{7}$$

Equivalently, $\alpha$ is the ratio of Coulomb energy to quantum energy on the same sphere, $(e^2/(4\pi\varepsilon_0 r))/(\hbar c/r)$, for every radius $r$ **[K]**. The same holds for the muon and the tau: their Coulomb and Compton spheres also stand in the ratio $\alpha$. $\alpha$ therefore does not distinguish the particles; it is a property of the charge unit, not of the mass **[B]**.

Geometrically, $\alpha = 1$ means that the Coulomb sphere, the Compton sphere and the Bohr radius merge into a single scale $1/m$. The three atomic lengths are not three independent scales but one scale with three powers of $\alpha$ (A267). Written out with $1/137$ this common origin stays hidden; the basic form exposes it. The SI value only says that in the historical charge unit the Coulomb sphere is 137 times smaller than the Compton sphere.

**Two levels.** The question about the 137 contains two different things. The first level is the normalisation: how one fixes the charge unit, $\varepsilon_0$, the $4\pi$ and $\hbar$. On this level "137" is only the name of the coupling. With $h$ instead of $\hbar$ the same coupling would read $e^2/(\varepsilon_0 hc) = 2\alpha \approx 1/68.5$, without the $4\pi$ it would read $e^2/(\varepsilon_0\hbar c) = 4\pi\alpha \approx 1/10.9$, and with $\alpha = 1$ and $e = \sqrt{4\pi}$ the number disappears as a constant altogether **[Q]**. The second level is the physical scale ratio: how the electric scale relates to the mechanical scale within the same system. Examples are $r_e/\lambda_C$, the binding energy of hydrogen relative to the rest energy of the electron, $\alpha^2/2 = 2.66 \times 10^{-5}$, and the ratio of two resistances, $Z_0/R_K = 2\alpha$. Such ratios come out the same in every system of units. They also do not change when $e$ is redefined, because both sides are rescaled by the same factor; with $\alpha = 1$ they appear in the ratio $e_{\text{hist}}/e_{\text{geo}}$ **[B]**. The two levels became mixed because $\alpha$ is historically defined through the charge. Since 2019 the SI has additionally moved the measured value into $\varepsilon_0 = e^2/(2\alpha hc)$, so that $\alpha$ appears to be a property of the charge or of the vacuum. The basic form separates the two levels: $\alpha = 1$ settles the normalisation, and the scale ratio is what a theory has to deliver. FFGFT delivers $\alpha^{-1} = 3700/27$ for it (Doc. 338, 394), 7.6 ppm from the measured value **[K]**.

## 4 The bridge in the basic form: $\xi$ as an area

FFGFT links $\alpha$ to $\xi$ through the energy scale $E_0 = \sqrt{m_e m_\mu}$, the geometric mean of the two lightest charged leptons. In the basic form, i.e. in T0 units with $\alpha = 1$ and without fractal correction, the bridge reads

$$\xi E_0^2 = 1 \quad \Longleftrightarrow \quad E_0^2 = \frac{1}{\xi} = 7500, \qquad E_0 = 86.60 \quad \textbf{[B]}. \tag{8}$$

**$\xi$ as an area.** Since the Compton radii are the reciprocals of the masses, $\lambda_e = 1/m_e$ and $\lambda_\mu = 1/m_\mu$, the bridge is equivalent to

$$\xi = \frac{1}{m_e m_\mu} = \lambda_e \lambda_\mu \quad \textbf{[B]}. \tag{9}$$

In the basic form $\xi$ is the product of the Compton radii of electron and muon, i.e. the area of the rectangle spanned by the two Compton spheres. Because with $\alpha = 1$ the Coulomb sphere of the electron coincides with its Compton sphere, one equally has $\xi = r_e \lambda_\mu$: the Coulomb sphere of the electron times the Compton sphere of the muon **[B]**. The bridge between charge and geometry is thus a statement about two sphere radii, without a unit and without a correction.

**The same number as the recursion.** The number $1/\xi = 7500$ splits as $7500 = 100 \cdot 75$. Exactly these two factors carry the time structure: the recursion $\xi_{n+1} = \xi_n(1 - 100\,\xi_n)$ starts at $\xi_0 = 1/7500$, the first increment is $100\,\xi_0 = 1/75$, and in the frozen case the winding closes after exactly 75 turns with the rotation number $74/75 = 1 - 100\xi$ (Docs. 381, 384) **[B]**. The number that fixes the energy unit of the basic form through $E_0^2 = 1/\xi$ is therefore the same one that determines the winding of the time structure. The value of the factor that later appears as $K_{\text{frak}}$ in the translation into SI is thus already present in the basic form, as geometry and not as a correction.

**The T0 energy unit.** Taken by itself, $\xi E_0^2 = 1$ fixes the energy unit of the basic form: one unit is $E_0/86.60 = \sqrt{\xi m_e m_\mu} = 84.85\,\text{keV}$ **[K]**. All statements of the basic form are ratios in this unit; the comparison with the MeV is the transition to SI.

# 5 The translation into SI

## The reference energy

In SI the ratio form becomes (Doc. 011, A130)

$$\alpha_{\text{SI}} = \xi\left(\frac{E_0}{E_{\text{ref}}}\right)^2, \qquad E_{\text{ref}} = 1\ \text{MeV}. \tag{10}$$

In SI, $\xi E_0^2$ has the dimension energy$^2$, so a reference energy is needed. Computed backwards, the measured value requires $E_{\text{ref}} = E_0\sqrt{\xi/\alpha} = 0.99323\,\text{MeV}$ with the bare $E_0 = 7.348\,\text{MeV}$, which is $\sqrt{K_{\text{frak}}}\,\text{MeV}$ to within $8 \times 10^{-5}$, and $0.99992\,\text{MeV}$ with the corrected $E_0 = 7.397\,\text{MeV}$ **[K]**. Without correction $\alpha^{-1} = 7500/E_0^2 = 138.9$, with it $137.06$, $1.7 \times 10^{-4}$ from the measured value. This agreement is a property of the anchor $E_0$ (R135), not an independent prediction **[K]**.

The area form of the bridge also has its SI counterpart: the classical electron radius times the Compton length of the muon is

$$r_e\,\lambda_\mu = \frac{\xi}{K_{\text{frak}}}\left(\frac{\hbar c}{1\ \text{MeV}}\right)^2 \quad (+1.7 \times 10^{-4}) \quad \textbf{[K]}, \tag{11}$$

$1.4\,\%$ off without correction. The area of the basic form appears in SI in units of $(\hbar c/\text{MeV})^2$, and it is exactly there that $K_{\text{frak}}$ enters.

## Where the MeV comes from

The MeV is not a geometric quantity but a unit of the SI chain **[Q]**: $1\,\text{MeV} = 10^6 \cdot 1.602176634 \times 10^{-19}\,\text{J}$. The numerical factor is the numerical value of the elementary charge in coulombs, fixed since 2019 and taken over from the old definition of the ampere with $\mu_0 = 4\pi \times 10^{-7}\,\text{N/A}^2$; the joule depends on the kilogram, metre and second. The MeV therefore carries the historical charge unit, exactly the unit in which $\alpha$ hides (Section 2). The electron mass in MeV also has a measurement path involving $\alpha$: what is measured is the Rydberg energy $\text{Ry} = 13.6057\,\text{eV}$, and $m_e c^2 = 2\,\text{Ry}/\alpha^2$ **[B]**. The SI form of the bridge is equivalent to $m_e/\text{MeV} = \sqrt{\alpha K_{\text{frak}}/(\xi m_\mu/m_e)} = 0.51104$, the same residual of $8 \times 10^{-5}$ **[K]**.

### Where the fractal correction sits

The two spheres show on which side the correction belongs. The Compton sphere contains only the mass, i.e. the anchor. The Coulomb sphere contains $e$, i.e. the same historical unit from which the MeV is built. If the correction is placed on this side, the masses and the Compton spheres stay bare, and the reference energy becomes $E_0^{\text{bare}}\sqrt{\xi/\alpha}/\sqrt{K_{\text{frak}}} = 0.99992\,\text{MeV}$. Mathematically this is the same equation as with the corrected $E_0$ (R72); the reading is more coherent, because $K_{\text{frak}}$ then sits on the side of the unit **[S]**.

The two routes to $\alpha$ in A130 differ only by this factor. The factor required by $\alpha$ is

$$K_\alpha = \xi m_e m_\mu \alpha^{-1} = \left(\frac{E_{\text{ref}}^{\text{bare}}}{1\ \text{MeV}}\right)^2 = 0.98650 \quad \textbf{[B]}, \tag{12}$$

$1.7 \times 10^{-4}$ from $74/75$ **[K]**; this is the same equation as in A130, read differently. In ratios $K_{\text{frak}}$ cancels (Doc. 122), and in the basic form $\xi E_0^2 = 1$ there is no room for it. In the $\alpha$ sector it therefore appears only in the translation into SI, as the adjustment between the T0 unit and the MeV **[K]**; its value $74/75$ comes from the geometry of the recursion (Section 4). The residual of $1.7 \times 10^{-4}$ remains open (R135, R139) **[S]**.

## 6 What holds and what is open

| Statement | Status | Reason |
|---|---|---|
| *Basic form (natural units, $\alpha = 1$)* | | |
| $e^2 = 4\pi\alpha$; with $\alpha = 1$, $e = \sqrt{4\pi}$, the full sphere | **[B]** | $\hbar = c = \varepsilon_0 = 1$ |
| $\alpha = (e_{\text{hist}}/e_{\text{geo}})^2$, calculation reversible | **[B]** | A267 |
| $r : \lambda_C : a_0 = \alpha : 1 : \alpha^{-1}$ for every charged particle | **[B]** | two spheres |
| $\alpha = 1$ as unit of calculation | **[Q]** | A267 |
| $\xi E_0^2 = 1$, $E_0^2 = 1/\xi = 7500$ | **[B]** | bridge |
| $\xi = \lambda_e\lambda_\mu = r_e\lambda_\mu$ | **[B]** | area of two spheres |
| $7500 = 100 \cdot 75$: $\xi_0 = 1/7500$, rotation number $74/75$ | **[B]** | recursion (Doc. 381) |
| T0 energy unit $\sqrt{\xi m_e m_\mu} = 84.85\,\text{keV}$ | **[K]** | check script |
| *Translation into SI* | | |
| $E_{\text{ref}} = 0.99992\,\text{MeV}$ (corrected), $0.99323\,\text{MeV}$ (bare) | **[K]** | property of the anchor (R135) |
| $r_e\lambda_\mu = (\xi/K_{\text{frak}})(\hbar c/\text{MeV})^2$ | **[K]** | $+1.7 \times 10^{-4}$ |
| MeV from the numerical value of $e$, $\mu_0 = 4\pi \times 10^{-7}$ | **[Q]** | SI chain |
| $m_e c^2 = 2\,\text{Ry}/\alpha^2$ | **[B]** | measurement path of the electron mass |
| correction on the unit side, Compton spheres bare | **[S]** | reading |
| $K_{\text{frak}}$ in the $\alpha$ sector only in the SI translation | **[K]** | basic form; Doc. 122 |
| residual $1.7 \times 10^{-4}$ between $K_\alpha$ and $74/75$ | **[S]** | R135, R139 |

## 7 Conclusion

With $\alpha = 1$ the basic form of FFGFT loses nothing and gains geometry. The charge is measured by the full surface of the sphere, $e = \sqrt{4\pi}$, and $\alpha$ becomes the charge unit: the ratio of Coulomb to Compton sphere, the same for every charged particle. With $\alpha = 1$ the

three atomic lengths merge into one scale. The bridge to $\xi$ then reads $\xi E_0^2 = 1$, which means geometrically: $\xi$ is the area spanned by the Compton spheres of electron and muon. The number $1/\xi = 7500 = 100 \cdot 75$ is at the same time the number of the recursion; the value $74/75$ is already present in the basic form. Only in the transition to SI does a reference energy come in, the MeV, built from the historical charge unit, and only there does the fractal correction act.

The same scale $1/m_e$ also carries gravitation through $\tilde{T} \cdot m = 1$ and $\xi$, $G \propto \xi^2/m_e$ (Doc. 012). In the static sector, that is for free fall, redshift and clocks, the mass reading carries this alone **[K]**. The full dynamics with light deflection, perihelion advance, frame dragging and gravitational waves is contained in the curvature representation through the field equations, whose form is enforced by Lovelock's theorem (Doc. 312); it remains open whether the structure of the carrier itself supplies the spatial half, the co-moving field and the tensor part (Doc. 308, 392) **[S]**. This gives a common scale framework for electrodynamics and gravitation, but not a completed unification.

*Check script:* `2/python/Dok386_Skripte/pruef_386_alpha_geometrie.py` — all numbers recomputed.